

Hand calculations for Problem 1

Input shape = (B, channels, height, width)

where B is batch size

$$= (B, 3, 32, 32)$$

$$\text{output size} = \left\lceil \frac{N - F + 2P}{S} \right\rceil + 1$$

Conv1: $F=3, S=1, P=1$, output channels = 32

$$\text{output size} = \left\lceil \frac{32 - 3 + 2}{1} \right\rceil + 1 = 32$$

$$\text{shape} = (B, 32, 32, 32)$$

Pool 1: $F=2, S=2, P=0$

$$\text{output size} = \left\lceil \frac{32 - 2 + 0}{2} \right\rceil + 1 = 16 \Rightarrow \text{Shape} = (B, 32, 16, 16)$$

Conv 2: $F=3, S=1, P=1$, output channels = 64

$$\text{output size} = \left\lceil \frac{16 - 3 + 2}{1} \right\rceil + 1 = 16 \Rightarrow \text{Shape} = (B, 64, 16, 16)$$

Pool 2: $F=2, S=2, P=0$

$$\text{output size} = \left\lceil \frac{16 - 2 + 0}{2} \right\rceil + 1 = 8 \Rightarrow \text{Shape} = (B, 64, 8, 8)$$

Conv 3: $F=3, S=1, P=1$, output channels = 128

$$\text{output size} = \left\lceil \frac{8 - 3 + 2}{1} \right\rceil + 1 = 8 \Rightarrow \text{Shape} = (B, 128, 8, 8)$$

Pool 3: $F=2, S=2, P=0$

$$\text{output size} = \left\lceil \frac{8 - 2 + 0}{2} \right\rceil + 1 = 4 \Rightarrow \text{Shape} = (B, 128, 4, 4)$$

Flatten:

channels $\times$ height $\times$ width

$$128 \times 4 \times 4 = 2048 \Rightarrow \text{Shape} = (B, 2048)$$